

Lab_Formal_3

You have to check the design equivalence of the generated netlist post **dft** Verses RTL Verilog files using Formal Verification Techniques.

Steps: -

1. Copy **Lab_Formal_3** folder from your computer into the virtual machine inside **Labs** directory
2. Open **dft_script.tcl**
 - Add the following command under section “**Formality Setup File**” to generate svf file used in formal verification
 - **set_svf ALU.svf**

```
6 ##### Define Working Library Directory #####
7
8 define_design_lib work -path ./work
9
10 ##### Formality Setup file #####
11
12 set_svf ALU.svf
13
14 ##### Design Compiler Library Files #setup #####
15
```

- Add the following command before write command under section “**Close Formality Setup File**” to stop writing in svf file and close it

```
136 ##### Close Formality Setup file #####
137
138 set_svf -off
139
140 #####
141 # Write out Design after initial compile
142 #####
143
144 #Avoid Writing assign statements in the netlist
145 change_name -hier -rule verilog
146 write_file -format verilog -hierarchy -output ALU_dft.v
147
```

3. Run synthesis using the following command to generate formality setup file “**SVF File**”

- **dc_shell -f dft_script.tcl | dft.log**

4. Open formality script and add the following: -

- don't verify constraints on scan input and output ports
- Contant value on all ATPG enable signals

5. Run the script in fm folder using the following command

- `fm_shell -f fm_script.tcl | fm.log`

6. Check that all the compare points are **succeeded** and there are no failing points